Curriculum Vitae

Rodrigo VEIGA

Full name: Rodrigo Soares Veiga.
Nationality: Brazilian.
Date of birth: May 2, 1988.
Professional address:
École Polytechnique Fédérale de Lausanne (EPFL);
School of Computer and Communication Sciences (IC);
Information Processing Group (IPG);
Lab for Statistical Mechanics of Inference in Large Systems (SMILS).
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Education

- **PhD, Physics**.
  University of São Paulo; São Paulo; Brazil.
  Thesis: *Statistical Physics Analysis of Machine Learning Models*.
  Date of Graduation: August 4, 2022.

- **Master of Sciences, Physics**.
  University of São Paulo; São Carlos; Brazil.
  Thesis: *Effects of Correlated Hybridization in the Single-impurity Anderson Model*.
  Date of Graduation: May 31, 2012.

- **Bachelor in Physics**.
  University of São Paulo; São Carlos; Brazil.
  Date of Graduation: December 21, 2009.

Research/Employment History

- **Postdoctoral researcher.**
  École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
  SMILS - Lab for Statistical Mechanics of Inference in Large Systems.
  Supervisor: Prof. Nicolas Macris.
  Dates: October, 2022 – (current job).

- **Visiting PhD student**.
  École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
  IdePHICS - Information, Learning and Physics Lab.
  Supervisor: Prof. Florent Krzakala.
  Dates: February, 2021 – January, 2022.

- **PhD student**.
  University of São Paulo (USP); São Paulo; Brazil.
  IFUSP - Institute of Physics.
  Supervisor: Prof. Renato Vicente.
  Dates: July, 2017 – August, 2022.

- **Financial administrator**.
  Primos Materiais para Construções Ltda; Sorocaba; Brazil.
  Non-academic employment.
  Dates: July, 2013 – March, 2017.

- **PhD student**.
  University of São Paulo (USP); São Carlos; Brazil.
  IFSC - São Carlos Institute of Physics.
  Supervisor: Prof. Miled Moussa.
  Dates: August, 2012 – May, 2013 (interrupted).

- **Master student**.
  University of São Paulo (USP); São Carlos; Brazil.
  IFSC - São Carlos Institute of Physics.
  Supervisor: Prof. Valter Líbero.
  Dates: March, 2010 – May, 2012.

- **Undergraduate student project**.
  University of São Paulo (USP); São Carlos; Brazil.
  IFSC - São Carlos Institute of Physics.
  Supervisor: Prof. Valter Líbero.
  Dates: April, 2008 – December, 2009.

Publications

- **Denoising Score Matching with Random Features: Insights on Diffusion Models from Precise Learning Curves**.
  Authors: A. J. George, **R. Veiga**, N. Macris.
  - Published: *Preprint* (2025). Under review.
  - arXiv: 2502.00336

- **Analysis of Diffusion Models for Manifold Data.**
  Authors: A. J. George, **R. Veiga**, N. Macris.
  - Published: *Preprint* (2025). Under review.
  - arXiv: 2502.04339

- **Stochastic gradient flow dynamics of test risk and its exact solution for weak features**.
  Authors: **R. Veiga**, A. Remizova, N. Macris.
  - Published: *Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR 235:49310-49344* (2024).
  - arXiv: 2402.07626

- **Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks**.
  Authors: **R. Veiga**, L. Stephan, B. Loureiro, F. Krzakala, L. Zdeborová.
  - Published: *Advances in Neural Information Processing Systems (NeurIPS). Volume 35; pages 23244-23255* (2022).
  - Re-published: *Journal of Statistical Mechanics; Theory and Experiment. Volume 2023, page 114008, DOI 10.1088/1742-5468/ad01b1* (2023).
  - arXiv: 2202.00293

- **Learning curves for the multi-class teacher–student perceptron**.
  Authors: E. Cornacchia*, F. Mignacco*, **R. Veiga***, C. Gerbelot, B. Loureiro, L. Zdeborová (*equal contribution).
  - Published: *Machine Learning: Science and Technology. Volume 4; pages 05019; DOI 10.1088/2632-2153/acb428* (2023).
  - arXiv: 2203.12094

- **Restricted Boltzmann machine flows and the critical temperature of Ising models**.
  Authors: **R. Veiga**, R. Vicente.
  - Published: *Preprint* (2020).
  - arXiv: 2006.10176

- **Age-structured estimation of COVID-19 ICU demand from low quality data**.
  Authors: **R. Veiga**, R. Murta, R. Vicente.
  - Published: *Preprint* (2020).
  - arXiv: 2006.06530

Participation in events

Oral presentations

- **The Abdus Salam International Centre for Theoretical Physics (ICTP); Trieste, Italy.**
  Contributed talk: *Time Evolution of the Test Risk under Stochastic Gradient Flow Dynamics*.
  Event: Youth in High Dimensions: Recent Progress in Machine Learning, High-Dimensional Statistics and Inference.
  Date: May, 2024.

- **African Institute for Mathematical Sciences (AIMS); Kigali, Rwanda**.
  Invited speaker: *Time Evolution of the Test Risk under Stochastic Gradient Flow Dynamics*.
  Event: From Theory to Practice: Workshop in Data Science.
  Date: April, 2024.

- **TOPML, Workshop on the Theory of Overparameterised Machine Learning; Houston, USA**.
  Contributed talk: *Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks.*
  Virtual event.
  Date: April, 2022.

Poster presentations

- **ICML, International Conference on Machine Learning; Vienna; Austria**.
  Poster title: *Stochastic Gradient Flow Dynamics of Test Risk and its Exact Solution for Weak Features*.
  Date: July, 2024.

- **Institut d'Études Scientifiques de Cargèse; Cargèse, France**.
  Poster title: *Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks.*
  Event: Statistical Physics and Machine Learning Back Together.
  Date: August, 2023.

- **NeurIPS, Conference on Neural Information Processing Systems; New Orleans, USA**.
  Poster title: *Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks.*
  Date: December, 2022.

- **University of São Paulo, São Carlos Institute of Physics; São Carlos, Brazil**.
  Poster title: *Entanglement and quantum Discord in the superradiance.*
  Event: II SIFSC - São Carlos Physics Institute Graduate Workshop.
  Date: October, 2012.

- **University of São Paulo, São Carlos Institute of Physics; São Carlos, Brazil**.
  Poster title: *Effects of correlated hybridization in the single-impurity Anderson model.*
  Event: I SIFSC - São Carlos Physics Institute Graduate Workshop.
  Date: October, 2011.

- **Federal University of Rio Grande do Norte, IIP; Natal, Brazil.**
  Poster title: *Effects of correlated hybridization in the single-impurity Anderson model.*
  Event: Brazilian School on Statistical Mechanics.
  Date: July, 2011.

- **University of São Paulo, São Carlos Institute of Physics; São Carlos, Brazil**.
  Poster title: *Correlated hybridization in the single-impurity Anderson model and non-local functional in the Heisenberg model.*
  Event: XIV São Carlos Physics Institute Graduate Workshop.
  Date: October, 2010.

Schools

- **ICTP South American Institute for Fundamental Research; São Paulo, Brazil.**
  Event: First School on Data Science and Machine Learning.
  Date: December, 2019.
- **ICTP South American Institute for Fundamental Research; São Paulo, Brazil.**
  Event: Minicourse on Machine Learning for Many-Body Physics.
  Date: September, 2017.

Scholarships

- **CAPES-PrInt, Program for Institutional Internationalization; Brazil.**
  Scholarship for a PhD internship abroad.
  Project: Statistical physics inference on machine learning algorithms.
  Grant number: 88887.467036/2019-00.
  Host: École Polytechnique Fédérale de Lausanne; Switzerland.
  Dates: February, 2021 – January, 2022.

- **CNPq, The National Council for Scientific and Technological Development; Brazil**.
  PhD scholarship.
  Project: Statistical physics and machine learning models.
  Grant number: 162857/2017-9.
  Dates: August, 2017 – February, 2020.

- **FAPESP, The State of São Paulo Research Foundation; Brazil**.
  PhD scholarship.
  Project: Entanglement and quantum discord in the superradiance and applications of quantum information theory in NMR.
  Grant number: 2012/12065-7.
  Dates: Declined.

- **CAPES, Coordination for the Improvement of Higher Education Personnel; Brazil**.
  PhD scholarship.
  Project: Entanglement and quantum discord in the superradiance and applications of quantum information theory in NMR.
  Grant number: PROEX.
  Dates: August, 2012 – May, 2013.

- **FAPESP, The State of São Paulo Research Foundation; Brazil**.
  Master scholarship.
  Project: Effects of correlated hybridization in the single-impurity Anderson model.
  Grant number: 2009/13065-8.
  Dates: March, 2010 – February, 2012.

- **FAPESP, The State of São Paulo Research Foundation; Brazil**.
  Undergraduate research scholarship.
  Project: *Density functional theory applied to the antiferrimagnetic Heisenberg model*.
  Grant number: 2007/59988-4.
  Dates: April, 2008 – December, 2009.

Teaching

- **CS526 Learning theory, EPFL; Lausanne; Switzerland**.
  Master's course whose main instructor is Prof. Nicolas Macris.
  Spring semester 2024
  Responsible for two lectures:
  - March 18, 2024: Bias variance tradeoff and the double descent phenomenon.
  - March 25, 2024: Double descent and the weak features model.
  Spring semester 2025
  - February 17, 2025: PAC learning framework and uniform convergence.
  - March 17, 2025: Bias variance tradeoff and the double descent phenomenon.
  - March 24, 2025: Double descent and the weak features model.

Academic service

- **Reviewer**
  - Physical Review E.
  - International Conference on Machine Learning (ICML).
  - IEEE International Symposium on Information Theory (ISIT).

Outreach project

- **Modeling growth rate and growth acceleration rate of COVID-19 cases in Brazil.**
  Code available on GitHub.
  Brazilian newspaper *Folha de S. Paulo* used this model to construct an acceleration monitor of COVID-19 in Brazil.
  - Several press reports followed from the acceleration monitor based on the project: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

References

- **Prof. Nicolas Macris.**
  École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
  IC - School of Computer and Communication Sciences.
  SMILS - Lab for Statistical Mechanics of Inference in Large Systems.
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- **Prof. Florent Krzakala.**
  École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
  STI - School of Engineering.
  IdePHICS - Information, Learning and Physics Lab.
  florent.krzakala@epfl.ch

- **Dr. Bruno Loureiro.**
  École Normale Supérieure - PSL & CNRS; Paris; France.
  Département d'Informatique.
  Center for Data Science.
  bruno.loureiro@di.ens.fr

- **Prof. Renato Vicente.**
  University of São Paulo (USP); São Paulo; Brazil.
  IME - Institute of Mathematics and Statistics.
  MAP - Applied Mathematics Department.
  rvicente@usp.br

- **Prof. Nestor Caticha.**
  University of São Paulo (USP); São Paulo; Brazil.
  IF - Institute of Physics.
  FGE - General Physics Department.
  ncaticha@usp.br